HHID: A Human-Human Interaction Dataset for Contact-Aware Novel View Synthesis

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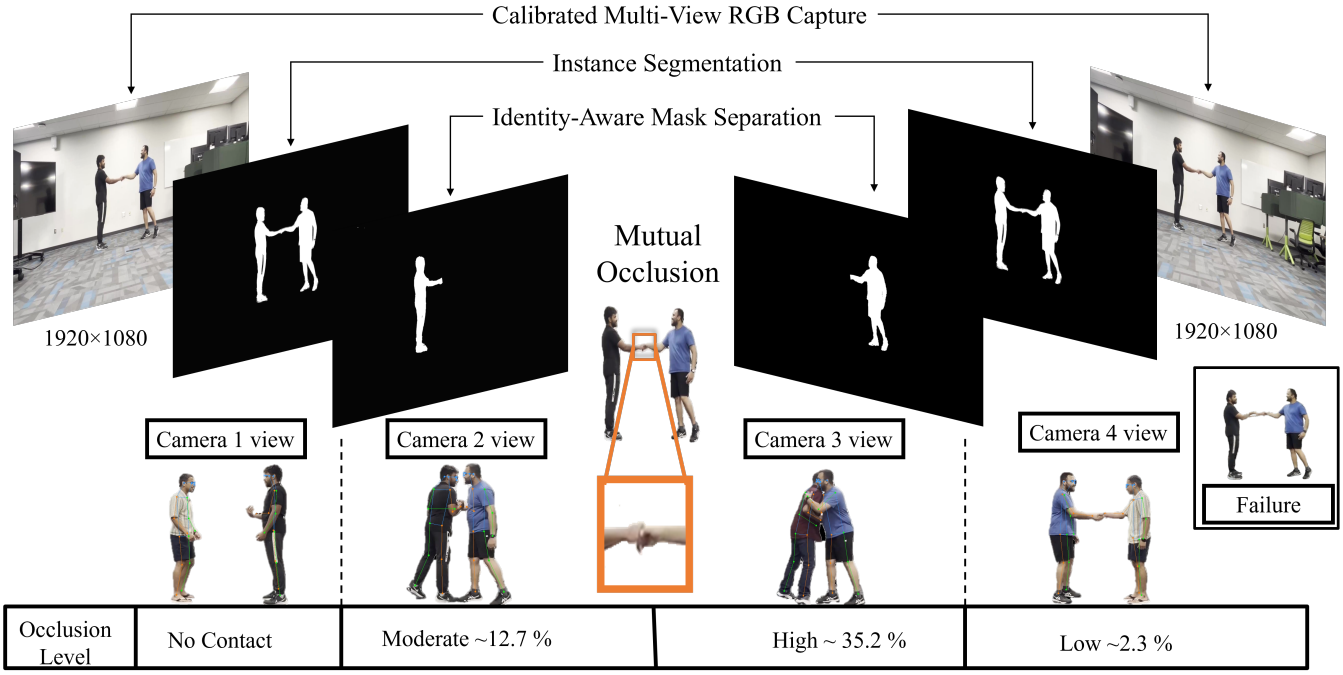


Figure 1: HHID captures multi-view two-person interactions featuring sustained contact and severe cross-occlusion. Example novel view synthesis (NVS) output under sustained cross-occlusion, illustrating geometric distortion.

Abstract

Recent advances in neural rendering have enabled high-quality novel view synthesis for single human subjects; however, existing methods remain limited in human-human interaction scenarios involving sustained physical contact and severe cross-occlusion. These conditions introduce identity ambiguity that is not adequately captured by current benchmarks. We present HHID, a human-human interaction dataset designed to evaluate identity-consistent novel view synthesis under sustained contact and cross-occlusion. The dataset consists of synchronized multi-view RGB sequences of two-person interactions captured with four calibrated cameras at 30 FPS, totaling approximately 1500 frames per camera. HHID provides rich identity-aware annotations, including foreground and per-subject instance masks, 2D keypoints, triangulated 3D joints, and temporally consistent SMPL parameters. We further define benchmark tasks and identity-aware evaluation protocols for contact-aware human interaction rendering and evaluate representative state-of-the-art methods. Experimental results demonstrate consistent performance degradation with increasing interaction intensity, revealing fundamental limitations in preserving identity and geometry under occlusion. HHID establishes a diagnostic benchmark for studying identity-consistent neural rendering in human-human interaction scenarios and enables future research on contact-aware dynamic scene understanding.

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CCS Concepts

• **Computing methodologies** → **Reconstruction**.

Keywords

human-human interaction, novel view synthesis, cross-occlusion, identity-consistent rendering

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1 Introduction

Neural rendering has enabled photorealistic synthesis of human appearance from novel viewpoints, driven by methods such as NeRF and Gaussian-based representations [8, 10, 18]. These approaches perform well in single-subject settings, where geometry and appearance can be modeled independently. However, real-world human interactions often involve close contact and cross-occlusion, where these assumptions break down. In such scenarios, subjects become spatially entangled, creating ambiguity in geometry, appearance, and identity, and leading to artifacts such as identity mixing, geometric distortion, and temporal inconsistency [19]. A key limitation is the lack of datasets that explicitly capture and evaluate these interaction-driven challenges. Existing human-centric benchmarks primarily focus on isolated subjects, human-object interaction, or motion generation [1, 5, 20, 27], while interaction datasets such as Hi4D and InterHuman are not designed as benchmarks for identity-consistent neural rendering under sustained cross-occlusion [14, 28]. As a result, current methods are neither trained nor systematically evaluated for preserving identity and geometric consistency under human-human interaction. Fig. 1 provides a visual overview of human-human interactions captured across synchronized views. As physical contact increases, subjects become spatially entangled, resulting in severe cross-occlusion and identity ambiguity that challenge existing neural rendering methods, often leading to identity mixing and geometric distortion.

1.1 Contributions

We make the following contributions:

- **HHID dataset.** We introduce HHID, a synchronized multi-view dataset of two-person interactions featuring sustained contact, severe cross-occlusion, and identity ambiguity.
- **Identity-aware annotations.** HHID provides calibrated cameras, per-subject instance masks, 2D keypoints, triangulated 3D joints, and temporally consistent SMPL parameters.
- **Benchmark and evaluation.** We establish standardized splits, benchmark tasks, and introduce the Identity Switching Rate (ISR) metric for evaluating identity consistency under severe occlusion.
- **Baseline evaluation.** We benchmark representative neural rendering methods and demonstrate the challenges of preserving geometry and identity during close human-human interaction.

The dataset and accompanying documentation are available for download at: https://github.com/sohomd/HHID

2 Related Work

2.1 Human-Centric Neural Rendering Datasets

Human-centric datasets have been central to neural rendering and dynamic human reconstruction. ZJU-MoCap [5] and NeuralBody [20] provide synchronized multi-view capture with accurate pose and geometry annotations for controlled single-subject synthesis. Monocular and sparse-view datasets such as People-Snapshot [1] and MonoPerfCap [27] focus on personalized performance capture under limited viewpoints. Larger-scale datasets, including THuman [32] and CAPE [17], expand geometric and clothing diversity, while AIST++ [13] introduces multi-view motion sequences. However, these benchmarks primarily focus on isolated subjects or loosely coupled motion, limiting their ability to model close interaction where independent geometry assumptions break down.

2.2 Multi-Person Capture Datasets

Multi-person datasets extend capture to interactive settings. Panoptic Studio [9] enables large-scale multi-person pose reconstruction, while Hi4D [28] provides two-person motion capture with parametric annotations. BEHAVE [2] focuses on human-object contact. In most cases, interaction duration and occlusion severity remain limited, and identity association is largely stable across views and time, reducing the need to resolve identity ambiguity under sustained contact. Recent dataset, InterGen [14] focuses on human-human motion generation rather than multi-view neural rendering and identity-aware evaluation.

2.3 Geometry and Scan Datasets

Geometry-focused datasets such as THuman4.0 [31] and CAPE [17] provide high-quality clothed human models with parametric supervision, while platforms such as Twindom [24] offer high-fidelity static scans. However, these datasets focus on static or single-subject capture and lack temporally continuous interaction. Consequently, they do not support evaluation under sustained contact, severe cross-occlusion, or identity ambiguity. Table 2 summarizes multi-person interaction datasets. Compared with existing benchmarks, HHID provides sustained human-human contact, occlusion duration, and introduces an identity-aware rendering benchmark through the proposed Identity Switching Rate (ISR).

3 HHID Dataset

3.1 Motivation

From the above discussion, we observe that existing human-centric datasets primarily focus on isolated subjects or loosely coupled motion [5, 27], and do not capture sustained human interaction under severe cross-occlusion. Current methods are neither designed nor systematically evaluated to resolve identity ambiguity when subjects overlap and become spatially entangled. To address this limitation, we introduce HHID, a human-human interaction dataset for contact-aware novel view synthesis under cross-occlusion and identity ambiguity. Furthermore, HHID focuses on two-person interactions as the fundamental unit of human-human contact, enabling controlled analysis of occlusion and identity ambiguity.

3.2 Capture Setup

Sequences are recorded in a controlled indoor environment using four synchronized RGB cameras at 30 FPS, arranged to provide overlapping views of a fixed interaction region. Four subjects form two interaction pairs across five captured sequences. Each sequence contains ~300 frames per camera, totaling ~1500 frames per camera. Interaction types include no contact, handshake, shoulder contact,

Table 1: Comparison of human-centric neural rendering datasets. Existing benchmarks support single-subject modeling or loosely coupled multi-person motion but do not systematically evaluate contact and identity entanglement under severe cross-occlusion. HHID is explicitly structured to address this evaluation gap.

Dataset	Multi-Person	Sustained Contact	Severe Occlusion	Identity Annotations	Dynamic Multi-View	SMPL Provided	Benchmark Protocol
ZJU-MoCap [5]	No	No	No	Yes	Yes	Yes	Yes
People-Snapshot [1]	No	No	No	Yes	No	Yes	No
MonoPerfCap [27]	No	No	No	Yes	No	Yes	No
AIST++ [13]	No	No	No	No	Yes	No	No
THuman2.0 [29]	No	No	No	No	No	Yes	No
CAPE [17]	No	No	No	No	No	Yes	No
Hi4D [28]	Yes	Limited	Limited	Yes	Yes	Yes	No
HHID (Ours)	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Table 2: Contact and occlusion statistics for HHID are measured using the proposed evaluation protocol. Values for the comparison datasets are estimated from representative interaction sequences.

Dataset	Interaction	People	Pairs	Cameras	Contact Frames	Max Contact Time (s)	Max Occl. Time (s)	Identity Metric	Rendering Benchmark
Panoptic Studio [9]	Human–Human	≤8/scene	–	521	~9	~0.4	~1.8	–	×
CHI3D [6]	Human–Human	10	5	4	~5	~0.6	~2.7	–	×
Harmony4D [11]	Human–Human	24	2/seq	20	~25	~2.5	~5.3	–	×
Hi4D [28]	Human–Human	40	20	8	~16	~1.5	~8.1	–	×
BEHAVE [2]	Human–Object	8	–	4 RGB-D	~22	~2.1	~4.2	–	×
HHID (Ours)	**Human–Human**	**4**	**2**	**4**	**~28**	**~5.10**	**~8.33**	**ISR**	✓

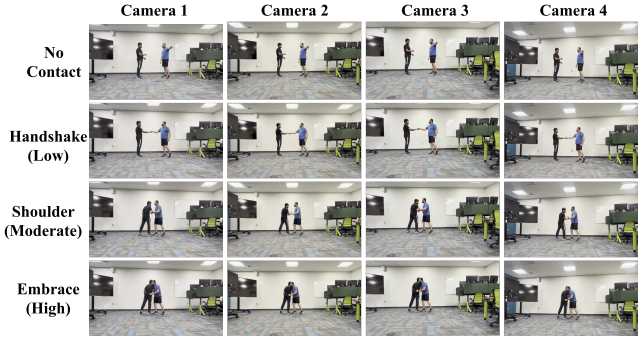


Figure 2: Overview of HHID dataset. Each row shows an interaction type, while columns correspond to synchronized views from four cameras.

and full-body embrace, representing increasing levels of proximity and occlusion. An overview of the dataset is shown in Fig. 2, illustrating the progression of interaction types and corresponding multi-view observations. The figure highlights increasing inter-subject proximity and occlusion across synchronized camera views.

3.3 Annotations

To enable identity-consistent evaluation under sustained cross-occlusion, HHID provides multi-level annotations generated through a structured and temporally coherent pipeline. All annotations are synchronized across views and indexed per subject identity.

Instance Segmentation and Identity Tracking. Per-frame instance segmentation masks are generated for each view [15], followed by identity-aware refinement. During severe overlap, manual verification is applied to maintain consistent subject labeling across frames and views, preventing identity switching under peak occlusion.

2D Keypoint Detection. For each subject in every view, 2D pose keypoints are extracted using a widely used 2D pose estimation method [3]. Keypoints are temporally indexed per identity to support consistent tracking.

3D Joint Reconstruction. Multi-view 2D detections are triangulated using calibrated camera parameters [22] to obtain 3D joint locations. Temporal smoothing is applied to reduce noise while preserving motion dynamics.

SMPL Parameter Estimation. Temporally consistent SMPL parameters are estimated using a learning-based framework [12], constrained by triangulated 3D joints and temporally regularized to prevent drift and reduce interpenetration.

Quality Control. All annotation layers are validated for cross-view alignment, temporal stability, and identity consistency, with particular emphasis on peak-contact intervals where occlusion ambiguity is highest. This multi-level annotation design enables joint evaluation of rendering fidelity, geometric accuracy, temporal coherence, and identity preservation in contact-heavy human-human interactions.

Table 3: Summary statistics of the HHID dataset.

Property	Value
Subjects	4
Interaction setting	Two-person interactions
Interaction categories	No contact, handshake, shoulder contact, embrace
Sequences	5 interaction sequences
Dataset split	70% / 10% / 20% (sequence-level)
Cameras per sequence	4 synchronized RGB cameras
Frame rate	30 FPS
Frames per interaction (per camera)	~300
Frames per camera (total)	~1.5K
Total multi-view RGB frames	~6,000 (4 views × ~1.5K)
Image resolution	1920×1080
Capture environment	Indoor calibrated setup
Camera calibration	Intrinsic and extrinsic parameters
Identity annotations	Persistent subject identities
Foreground masks	Per-frame, per-view
Instance masks	Two identities per frame/view
2D keypoints	Per subject, frame, and view
3D joints	Per subject and frame
SMPL parameters	Per subject with temporal consistency

3.4 Dataset Statistics

All sequences are recorded at spatial resolution of 1920×1080, ensuring full-body visibility across viewpoints. Each frame contains exactly two interacting subjects observed from four synchronized views. A summary of dataset properties and annotation modalities is provided in Table 3. The dataset comprises approximately 1500 frames per camera (around 6000 multi-view RGB frames in total), with each interaction sequence spanning approximately 300 consecutive frames at 30 FPS. Interaction types include no contact, handshake, shoulder contact, and full-body embrace, introducing controlled variation in spatial proximity and occlusion severity. To quantify interaction density, we define an inter-subject overlap ratio as the pixel-wise intersection-over-union (IoU) between per-subject instance masks. The dataset exhibits sustained periods of mutual occlusion, with peak-contact intervals showing significant mask overlap. This distribution enables systematic evaluation of rendering fidelity and identity consistency under interaction-heavy conditions.

An illustration of the SMPL parameterization used for per-subject modeling is shown in Fig. 3. Each subject is modeled per frame using a shared parametric body model with 6890 vertices, 24 kinematic joints, and 72-D pose parameters, enabling identity-consistent geometric and motion representation under interaction.

3.5 Data Organization

The HHID dataset is organized into modular components, including multi-view RGB images, union foreground masks, per-subject instance masks, 2D and 3D keypoints, SMPL parameters, and calibrated camera parameters. Annotations are indexed per frame and per subject identity. Structured metadata and predefined training, validation, and testing splits are provided to ensure reproducible benchmarking.

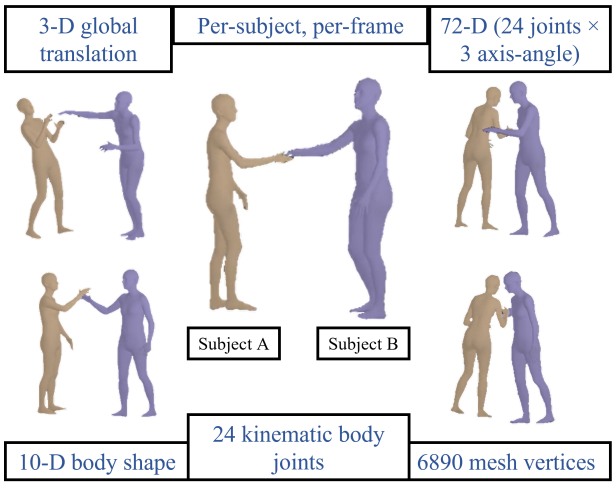


Figure 3: SMPL representation for two interacting subjects.

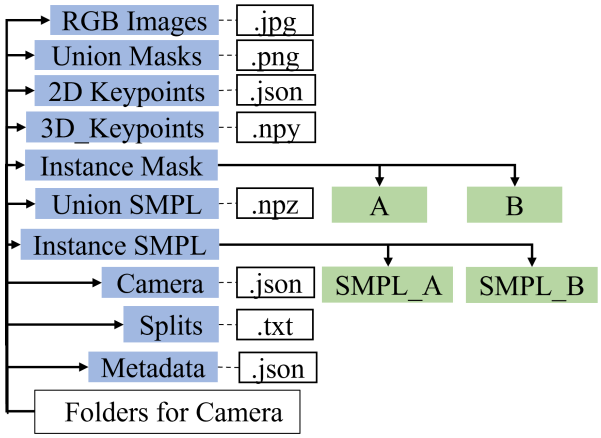


Figure 4: Directory structure for each camera sequence.

Fig. 4 illustrates the dataset structure. Multi-view RGB images are paired with identity-aware annotations, including union and instance masks, 2D/3D keypoints, and temporally consistent SMPL parameters. Camera calibration files provide intrinsic and extrinsic parameters for geometric consistency across views. Standardized splits and metadata support controlled evaluation under varying interaction and occlusion conditions.

3.6 Occlusion and Contact Statistics

We quantify interaction difficulty using an overlap ratio defined as the pixel-wise intersection-over-union (IoU) between per-subject instance masks. Occlusion severity increases from mild interactions to high-contact scenarios, enabling structured evaluation across low, moderate, and high-overlap conditions. A high-contact subset is defined for frames with an overlap ratio exceeding 30%, enabling targeted evaluation of rendering robustness, temporal stability, and identity preservation under severe inter-subject occlusion.

$$\text{IoU} = \frac{|M_A \cap M_B|}{|M_A \cup M_B|} \times 100 \tag{1}$$

Table 4: Occlusion statistics across interaction types in HHID. Values are reported as mean ± standard deviation over all frames.

Interaction Type	Mean (%)	Peak (%)	Contact Level
No contact	1.7 ± 0.1	~ 5	Low
Handshake	8.5 ± 3.2	~ 15	Low
Shoulder Contact	21.4 ± 6.8	~ 38	Moderate
Full-body Embrace	34.6 ± 9.3	~ 55	High

where M_A and M_B denote per-subject instance masks. The metric reflects the degree of mutual visibility loss due to spatial overlap. We categorize occlusion severity into three regimes:

- Low contact: < 10%
- Moderate contact: 10% – 30%
- High contact: > 30%

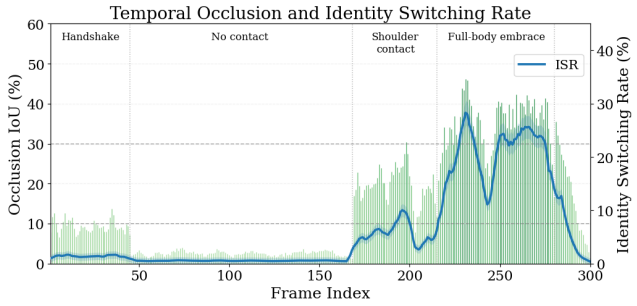


Figure 5: Temporal evolution of inter-subject occlusion (left axis), averaged over four synchronized camera views, and the corresponding Identity Switching Rate (ISR, right axis) across a representative interaction sequence.

As shown in Table 4, occlusion severity increases with interaction intensity, from limited overlap in low-contact scenarios to sustained high occlusion during shoulder contact and full-body embrace. This structured variation enables controlled evaluation under increasing identity ambiguity and visibility degradation. Occlusion is also pose-dependent, with upper-body joints more frequently occluded during contact, while lower-body joints remain comparatively visible. Figure 5 illustrates the temporal evolution of occlusion, where each vertical bar represents the mean IoU between subject masks averaged across views. The sequence captures transitions from low to high-contact interactions.

3.6.1 Identity Switching Rate. While overlap statistics quantify the severity of inter-subject occlusion, they do not directly measure whether a rendering method preserves subject identities during close interactions. We therefore introduce the *Identity Switching Rate (ISR)*, which measures the proportion of frames with incorrect identity assignment.

Let the binary variable s_t be defined as

$$s_t = \begin{cases} 1, & \text{if an identity switch occurs at frame } t, \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

where an identity switch is detected if

$$\begin{aligned} &\mathrm{IoU}(P_A^t, G_B^t) > \mathrm{IoU}(P_A^t, G_A^t) \\ \text{or} \quad &\mathrm{IoU}(P_B^t, G_A^t) > \mathrm{IoU}(P_B^t, G_B^t). \end{aligned} \tag{3}$$

Table 5: 2D visibility and 3D reconstruction statistics across interaction levels. Vis.=Visible (%), Occ.=Occluded (%), Err.=Mean error, Std.=Standard deviation.

Contact	2D Vis.	2D Occ.	3D Err. (mm)	Std (mm)
No	96.5	3.5	4.2	1.1
Low	89.2	10.8	6.8	1.9
Moderate	76.8	23.2	9.5	2.8
High	61.3	38.7	13.7	3.9

Here, P_A^t and P_B^t denote the predicted instance masks for subjects A and B, while G_A^t and G_B^t denote the corresponding ground-truth identity masks at frame t. For methods with subject-specific representations, predicted masks are obtained by rendering each subject independently. For scene-level methods that produce a single RGB rendering, per-subject masks are obtained by applying instance segmentation to the rendered output and associating the resulting regions with the ground-truth identities using IoU matching. The ISR is computed as

$$\mathrm{ISR} = \frac{1}{T} \sum_{t=1}^{T} s_t, \tag{4}$$

where T is the total number of evaluated frames. ISR is reported as the percentage of evaluated frames with incorrect identity assignment, where lower values indicate better identity preservation.

Figure 5 illustrates the relationship between inter-subject occlusion and ISR. As overlap increases, ISR also increases, illustrating how the proposed metric captures identity ambiguity under severe occlusion.

3.7 2D and 3D Keypoint Statistics

HHID provides dense 2D and 3D pose annotations for each subject across all views and frames. Following the COCO format [16], each frame contains $2 \times 4 \times 17$ annotated 2D keypoints for two subjects observed across four synchronized camera views. Keypoints are represented by pixel coordinates and confidence scores, and are extracted using a widely used pose estimator [3]. Corresponding 3D joints are triangulated using calibrated camera parameters, temporally aligned, and smoothed to improve consistency. While most joints remain visible under low-contact interactions, severe occlusion during close contact primarily affects upper-body joints, increasing uncertainty in both 2D detections and reconstructed 3D joints.

As shown in Table 5, keypoint visibility decreases while reconstruction error increases with interaction intensity. This structured degradation highlights the impact of cross-occlusion on pose estimation and geometric reconstruction, making HHID well-suited for evaluating pose-aware and geometry-consistent rendering.

4 Benchmark Tasks and Evaluation Protocol

HHID defines standardized benchmark tasks for evaluating contact-aware novel view synthesis under varying levels of inter-subject occlusion. The protocol is designed to assess rendering fidelity, identity preservation, geometric accuracy, and temporal consistency in interaction-heavy scenarios.

Table 6: Comparison of representative neural rendering methods on HHID. Results are reported over all frames and the high-contact subset, with ISR evaluating identity consistency.

Category	Method	All Frames (Mean)				High-Contact Subset			
		PSNR ↑	SSIM ↑	LPIPS ↓	ISR ↓	PSNR ↑	SSIM ↑	LPIPS ↓	ISR ↓
NeRF-based	NeRF [18]	23.8	0.812	0.214	18.5	20.1	0.741	0.278	32.1
NeRF-based	HumanNeRF [26]	26.3	0.861	0.162	12.8	22.4	0.793	0.221	25.7
Gaussian Splatting	3DGS [10]	27.8	0.889	0.138	10.2	23.6	0.821	0.198	21.3
Gaussian Splatting	Splatfacto (Nerfstudio) [23]	28.2	0.895	0.131	9.4	24.1	0.835	0.185	19.8
Gaussian Splatting	3DGS-Avatar [21]	29.1	0.907	0.118	7.8	25.3	0.854	0.162	17.2
Gaussian Splatting	GauHuman [8]	**29.8**	**0.915**	**0.105**	**6.5**	**26.0**	**0.867**	**0.149**	**15.2**
Human-Centric	NeuralBody [20]	26.9	0.872	0.151	11.6	22.8	0.804	0.209	23.5
Human-Centric	VIBE + Rendering [12]	24.9	0.832	0.192	15.2	21.2	0.761	0.259	28.4
Hybrid	TensoRF [4]	26.5	0.868	0.156	12.4	22.7	0.799	0.214	24.6
Hybrid	K-Planes [7]	27.4	0.882	0.141	10.9	23.5	0.818	0.201	22.1

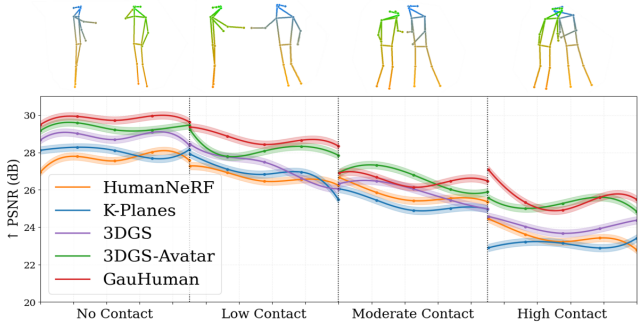


Figure 6: Temporal PSNR variation across interaction intensity, from no contact to high contact. All methods degrade with increasing occlusion.

4.1 Dataset Splits

The dataset is partitioned into training, validation, and testing sets (70%/10%/20%) at the interaction-sequence level to prevent temporal leakage. Although subjects appear across splits, interaction sequences are disjoint. A high-contact subset (IoU > 30%) is defined for evaluating performance under severe cross-occlusion.

4.2 Evaluation Metrics

Novel view synthesis quality is evaluated using PSNR, SSIM [25], and LPIPS [30]. Additional metrics include 3D joint error, temporal stability, and the proposed Identity Switching Rate (ISR), which measures the percentage of frames with incorrect identity assignment based on mask overlap. Lower ISR indicates better identity preservation. Mask-based metrics are computed independently for each camera view using manually verified identity-aware instance masks.

Fig. 6 illustrates the temporal variation of PSNR across increasing interaction intensity. All methods exhibit a consistent decline in reconstruction quality as occlusion increases, highlighting the impact of sustained interaction on rendering performance. While Gaussian-based methods such as GauHuman [8] and 3DGS-Avatar [21] demonstrate comparatively higher robustness, they still show noticeable degradation under high-contact conditions.

4.3 Benchmark Tasks and Baseline Experiments

HHID supports three benchmark tasks for interaction-aware rendering in human-human interaction scenarios: (i) **Multi-Human Novel View Synthesis:** Synthesize photorealistic images from unseen viewpoints while preserving geometric and visual consistency. (ii) **Identity-Consistent Rendering Under Occlusion:** Maintain subject identity during prolonged overlap and cross-occlusion. (iii) **Contact-Aware Dynamic Reconstruction:** Reconstruct temporally coherent geometry under sustained interaction, where single-subject assumptions break down.

Table 6 compares representative neural rendering methods on HHID. Although GauHuman [8] achieves the best overall performance, all methods degrade under severe interaction. While photometric metrics decrease with increasing occlusion, ISR exposes substantial identity switching, demonstrating the challenge of preserving geometry and identity during sustained human-human contact.

4.4 Limitations

HHID is captured in a controlled indoor environment with a limited number of subjects, which may restrict generalization to diverse real-world conditions. However, the dataset is intentionally designed as a diagnostic benchmark for evaluating identity consistency under sustained cross-occlusion and limited viewpoints. Unlike large-scale datasets that rely on automated annotations, HHID provides high-precision, manually verified identity labels during peak interaction. The dense temporal sampling enables statistically meaningful evaluation of spatial entanglement and identity ambiguity.

5 Conclusion

We introduced HHID, a human-human interaction dataset for evaluating contact-aware novel view synthesis under occlusion and identity ambiguity. The dataset provides synchronized multi-view capture, identity-aware annotations, and standardized evaluation protocols. Results show consistent degradation with increasing contact, highlighting limitations of existing methods. HHID establishes a diagnostic benchmark and introduces the Identity Switching Rate (ISR) metric for interaction-aware neural rendering.

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